

Zichen Zhang

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EDUCATION

M.S. in Applied Business Analytics (GPA:3.57)

Boston University, Metropolitan College, Boston, MA

Sep. 2025 – Dec.2026

B.S in Mathematics for Data Science

North China University of Technology and Brunel University London

Data Science and Big Data Technology Honorary Bachelor of Science Degree

Sep. 2021 – Jul. 2025

GPA: Second class(upper division) honours

INTERNSHIPS / EXPERIENCE

Data Analyst at *TSINO-TEK Beijing Co. Ltd.*

Aug. 2024 – Sep. 2024

- Extracted relevant raw data from the SMT database of the product line and reorganized the structure of the database in a new summative MS Excel document used for downstream data analysis.
- Conducted data preprocessing and used MATLAB to compare product dispatch columns with existing product production data; revealed a major difference between these numbers and reported this issue to management, which ultimately resolved it before damage was done.
- Organized data of customer orders of over 120 products into spreadsheets, used MATLAB formulas to time the requested number of product sets with the number of products in each set to obtain the actual demand of each product, and paralleled this data with inventory data to determine mismatches and prevent supply chain issues.

Full-Time Software Front-end Engineer at *Open Bayes*

Jan. 2024 - Mar. 2024

- Learned about JAVA and other programming knowledge from coding websites and practiced solving ~20 problems on a daily basis in the first two weeks of tenure.
- Collected and integrated data for the company's large-scale models by downloading training data from Wikipedia with a Linux terminal connected to the company's server.
- Converted data format with the bzip2 command to decompress data packages; processed data in the QA (Question-Answering) Dataset for clarity and accuracy and provided detailed documentation of the sources, format, and any special processing steps.

PROJECTS

Tesla ADAS Performance Analytics (FCW/AEB System Evaluation)

Student Consultant — Boston University Open Avenues Foundation & Forensic Rock, 2025

Sep. 2025 – Nov. 2025

- Collected and validated test measurements using **VBOX** high-precision automotive logging equipment; extracted speed, TTC, acceleration, and trigger-distance data for further analysis.
- Built comparative visual analytics showing FCW consistently triggers **~0.6 s and 43 ft earlier** than AEB, confirming reliable sequential warning-to-braking coordination.
- Quantified system robustness by analyzing avoidance success rates (**75%–84%**) and identifying degradation under **75% pedestrian offset** tests, highlighting detection limitations at higher lateral displacement.
- Evaluated braking patterns and found deceleration increases with speed ($-0.74\text{ g} \rightarrow -0.81\text{ g}$), indicating stronger braking under challenging conditions.
- Communicated analytical insights and recommended adjustments to improve offset detection and AEB trigger thresholds for enhanced crash mitigation performance.

Regional Data Collection and Ecological Investigation of Water Quality

Social Investigator and Data Analyst

Jul. 2022 – Aug. 2022

- Conducted field research at Miyun Water Reservoir, surveying 500+ participants (436 valid responses) and interviewing ~10 local residents regarding water quality and ecological environment perceptions.
- Visualized survey and national water-resource distribution using MATLAB (line charts, pie charts, histograms) and R (cascading color-coded geographic maps).
- Identified a positive correlation between perceived government intervention in water treatment and public opinion of environmental quality; found insufficient evidence linking rising environmental awareness to perceived water quality.
- Presented results to Miyun District government and CPPCC inspectors; research achieved **First Prize (Beijing Municipal Youth Service National Honor Roll)**, university-level **Project Excellence Award**, 2000+ online views, and media coverage from multiple regional outlets.